



What Is Evolution?

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Evolution = the change in the inherited traits of a species over many generations. It happens to a whole **population**, slowly — not to one individual in its lifetime. Every dog breed descended from wolves; that shift is evolution.

⚠️ **Individuals don't evolve — populations do.** One rabbit can't become more camouflaged in its life, but a rabbit population can over generations.

Genetic Variation

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Variation = the small differences in traits among members of a species. It is the raw material evolution works with. It comes from the **mixing of genes** in sexual reproduction and sometimes from **mutations** (changes in DNA).

💡 If every member were identical, one change could wipe them all out. Variation means some individuals likely have a trait that helps them survive a new challenge.

Natural Selection

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Natural selection = organisms with helpful traits survive & reproduce more, so those traits become common. **Charles Darwin** described it in **1859**.

1. Variation

Individuals differ in traits

2. Competition

More born than can survive

3. Selection

Helpful traits survive & reproduce

4. Inheritance

Survivors pass on the trait

★ Beetle story: light + dark beetles on dark soil → birds eat the easy-to-see light ones → after generations, most beetles are dark.

Survival of the Fittest

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Fitness = how well an organism is suited to survive & reproduce — NOT being strongest or fastest.

Trait	Why it raises fitness
Camouflage	Hides from predators, so it lives to reproduce
Sharp beak shape	Eats available food more easily
Disease resistance	Survivors pass on resistance
Bright mating colors	Attracts mates → more offspring

💡 **Peppered moths:** soot darkened trees → dark moths hid better → became the majority. When pollution dropped, light moths returned. Natural selection in real time.

Variation & the Environment

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The **environment decides** which traits are helpful — the same trait can help in one place and hurt in another.

Environment	Trait favored
Snowy arctic	White fur (camouflage)
Hot desert	Large ears (release heat)
Dark cave	No eyesight (saves energy)
Windy island	Smaller wings (not blown away)

★ A white coat saves an arctic hare but dooms a hare in a green forest. The environment makes a trait "good" or "bad."

The Math & the Fossil Record

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The Math of Selection

When a trait helps, its % of the population climbs each generation:

Gen	% Dark mice
1	30%
2	48%
3	66%
4	81%

The Fossil Record

A **fossil** is preserved remains/trace of past life. Deeper **rock layers** are older (**relative age**). The record shows:

- **Existence** — many organisms lived in the past
- **Diversity** — life grew more varied over time
- **Extinction** — many species died out

Fossils link ancient to modern life (a **common ancestor**).

Quick Check — Evolution

A. Put the 4 steps of natural selection in order: Inheritance, Variation, Selection, Competition.

_____ → _____ → _____ → _____

B. Does an individual or a population evolve? _____

C. In a snowy place, would white fur or brown fur be favored? _____

I Can... (Self-Check)

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- ☐ Define evolution as change over generations
- ☐ Explain why populations (not individuals) evolve
- ☐ Describe where variation comes from
- ☐ List the 4 steps of natural selection

- ☐ Explain "fitness" correctly
- ☐ Use the peppered moth as evidence
- ☐ Read how a trait's % climbs each generation
- ☐ Describe patterns in the fossil record

 Based on Georgia Standards of Excellence: S7L5 — You're ready for the unit test! 